

## Supporting Information

### **Kinetics of calcite nucleation onto sulfated chitosan derivatives and implications for water-polysaccharide interactions during crystallization of sparingly soluble salts**

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**Table S1:** Summary of Selected Studies of CaCO<sub>3</sub> Nucleation onto or in the Presence of Sulfur-Containing Macromolecules

| Material                         | CaCO <sub>3</sub> Polymorph and Morphology                 | Method and Notes                                                                                                                                                                                  | Ref. |
|----------------------------------|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <b>Polysaccharides</b>           |                                                            |                                                                                                                                                                                                   |      |
| Dermatan sulfate                 | Calcite, small columnar                                    | Observational, gas diffusion method, crystals were generally smaller and more numerous than with no treatment                                                                                     | 1    |
| Dermatan sulfate (de-sulfated)   | Calcite, rounded rhombohedral                              |                                                                                                                                                                                                   |      |
| Heparin                          | Rosette aggregates, polymorph unspecified                  |                                                                                                                                                                                                   |      |
| Heparin (low sulfation)          | Calcite, twinned                                           |                                                                                                                                                                                                   |      |
| Heparin                          | Calcite, rough rhombs                                      | Observational, ammonium carbonate method, crystals became porous as time increased from 1 day to 7 days                                                                                           | 2    |
| Heparin (low molecular weight)   | 30:70 calcite:vaterite, Irregular rhombs and spheres       | Observational, direct mixing, crystal size decreases and deformation increases with concentration of heparin                                                                                      | 3    |
| Heparin                          | CaCO <sub>3</sub> , polymorph unspecified                  | Observational, direct mixing, pH 9.75, authors suggest neutral polymers stabilize pre-nucleation clusters and charged polymers also stabilize but can't determine between pre and post nucleation | 4    |
| Dextran sulfate                  | CaCO <sub>3</sub> , polymorph unspecified                  |                                                                                                                                                                                                   |      |
| Keratan sulfate (avian derived)  | ACC <sup>1</sup> , vaterite (later transformed to calcite) | Quantitative; potentiometric titration, stabilization of prenucleation clusters observed, greater inhibition by keratan sulfate than dermatan sulfate                                             | 5    |
| Dermatan sulfate (avian derived) | ACC <sup>1</sup>                                           |                                                                                                                                                                                                   |      |
| Chondroitin-4-sulfate            | Vaterite, 77%<br>Calcite, 23%<br>Spheres and rhombs        | Observational, direct mixing                                                                                                                                                                      | 6    |
| Chondroitin-4-sulfate            | Vaterite                                                   | Quantitative, constant composition method,<br>$\gamma = 46 \text{ mJ m}^{-2}$<br>$\gamma = 62 \text{ mJ m}^{-2}$<br>$\gamma = 49 \text{ mJ m}^{-2}$                                               | 7    |
| Chondroitin-6-sulfate            | Vaterite                                                   |                                                                                                                                                                                                   |      |
| Dermatan sulfate                 | Vaterite                                                   |                                                                                                                                                                                                   |      |
| $\kappa$ -carragenan             | Calcite, intermediate between stacked and rosette crystals | Observational, direct mixing, compared to other polysaccharides, morphology based on if material formed a gel                                                                                     | 8    |
| $\kappa$ -carragenan             | Calcite, porous rhombs                                     | Observational, direct mixing with a double jet reactor (constant flow in but not out), all carrageenan materials increased induction and crystallization time                                     | 9    |
| $\iota$ -carragenan              | Calcite, porous rhombs, faceted                            |                                                                                                                                                                                                   |      |
| $\lambda$ -carragenan            | Calcite, porous rhombs, faceted                            |                                                                                                                                                                                                   |      |
| Dextran                          | Aragonite needles                                          | Observational, direct mixing method, pH 6                                                                                                                                                         | 10   |
| Dextran sulfate                  | Calcite and vaterite spheres                               | Phase separation of polymer observed                                                                                                                                                              |      |

|                                                                                                                                                                              |                                                                       |                                                                                                                                                                                                                |       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Polysaccharide from <i>E. Huxleyi</i> coccolith (sulfated and carboxylated)                                                                                                  | Calcite, induction time inhibited compared to solution alone          | Observational, direct mixing/ titration, hypothesize sulfate groups are in the interior of the molecule rather than the surface or bound to cations, conclude carboxyl groups play a larger role in inhibition | 11    |
| Polysaccharide from <i>E. Huxleyi</i> coccolith (95% desulfated)                                                                                                             | Calcite, induction time inhibited, no change from control             |                                                                                                                                                                                                                |       |
| Polysaccharide from <i>E. Huxleyi</i> coccolith (80% decarboxylated)                                                                                                         | Calcite, no inhibition                                                |                                                                                                                                                                                                                |       |
| Chitosan/ ι-carrageenan composite                                                                                                                                            | CaCO <sub>3</sub> , polymorph unspecified                             | Observational, direct mixing, composite dipped in biopolymer solution then immersed sea water (SSW)                                                                                                            | 12,13 |
| chitosan/ ι-carrageenan with PAA <sup>2</sup>                                                                                                                                | CaCO <sub>3</sub> , polymorph unspecified                             | Rates increased relative to control                                                                                                                                                                            |       |
| chitosan/ ι-carrageenan with citric acid                                                                                                                                     | CaCO <sub>3</sub> , polymorph unspecified                             | Rates increased relative to PAA <sup>2</sup> (and control)                                                                                                                                                     |       |
| chitosan/ ι-carrageenan with glutamic acid                                                                                                                                   | CaCO <sub>3</sub> , polymorph unspecified                             | Suggest acidity increases crystallization rate, glutamic>citric>PAA                                                                                                                                            |       |
| Chitosan and chondroidin-4-sulfate                                                                                                                                           | Vaterite, 77%<br>Calcite, 23%<br>Rough spheres                        | Observational, direct mixing, suggest chain rigidity influences morphology                                                                                                                                     | 6     |
| Proteins + protein/ polysaccharide mixtures                                                                                                                                  |                                                                       |                                                                                                                                                                                                                |       |
| Eggshell proteins of <i>G. domesticus</i> (chicken)                                                                                                                          | Calcite, columnar, fused                                              | Observational, <i>in vivo</i> , untreated                                                                                                                                                                      | 14    |
| Sulfation of eggshell proteins inhibited                                                                                                                                     | Calcite, spherical + short, poorly fused                              | Observational, <i>in vivo</i> , sodium chlorate treatment to inhibit sulfation of eggshell proteins                                                                                                            |       |
| Demineralized eggshell membranes of <i>G. domesticus</i> (chicken) containing sulfur residues                                                                                | Calcite rhombs associated with sulfur containing areas                | Observational, ammonium carbonate method                                                                                                                                                                       | 15    |
| Demineralized eggshell membranes of <i>G. domesticus</i> (chicken) with sulfur removed                                                                                       | Calcite, random distribution                                          |                                                                                                                                                                                                                |       |
| Demineralized eggshell membranes of <i>G. domesticus</i> (chicken) containing sulfur residues and crude organic matrix extract from <i>G. domesticus</i> (chicken)           | Calcite, modified and smaller than with eggshell membrane alone       | Observational, ammonium carbonate method, crystals localized around sulfur "patches"                                                                                                                           |       |
| Demineralized eggshell membranes of <i>G. domesticus</i> (chicken) containing sulfur residues and isolated dermatan sulfate proteoglycan from <i>G. domesticus</i> (chicken) | Calcite, modified, smaller and rounder than with crude organic matrix |                                                                                                                                                                                                                |       |
| Acidic proteins from <i>M. californianus</i> (bivalve)                                                                                                                       | Calcite, 14.6% of total crystallites                                  | Observational, ammonium carbonate diffusion                                                                                                                                                                    | 16    |

|                                                                      |                                                                     |                                                                                                                                                                     |       |
|----------------------------------------------------------------------|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
|                                                                      | oriented onto substrate                                             |                                                                                                                                                                     |       |
| Proteins from <i>U. pictorum</i> (bivalve)                           | Calcite aggregates, delayed nucleation                              | Observational, gas diffusion method, protein highly sulfated, complete inhibition at higher concentrations of protein                                               | 17    |
| Deglycosylated proteins from <i>U. pictorum</i> (bivalve)            | Calcite, deformed, no inhibition                                    |                                                                                                                                                                     |       |
| Nautilin-63, glycoprotein of <i>N. macromphalus</i> (mollusk)        | Calcite, deformed                                                   | Observational, gas diffusion method, Nautilin-63 is carboxylated and sulfated, deformation increased with concentration of protein                                  | 18    |
| Chitin and soluble OM from <i>A. Psittacus</i> (barnacle)            | Vaterite that transformed to rounded calcite, then rhombohedral     | Observational, gas diffusion method, deproteinization of chitin led to some deacetylation                                                                           | 19    |
| Isolated proteoglycans from <i>A. Psittacus</i> (barnacle)           | Calcite, modified, piles with c-axis alignment                      |                                                                                                                                                                     |       |
| Insoluble OM (containing chitin) from <i>A. Psittacus</i> (barnacle) | Calcite with c-axis orientation                                     | Observational, direct mixing, insoluble organic matrix contains proteoglycans                                                                                       | 20    |
| Soluble OM from <i>A. Psittacus</i> (barnacle)                       | Calcite, altered and aggregates                                     | Soluble organic matrix contains non-specific sulfates                                                                                                               |       |
| rOC90 <sup>3</sup>                                                   | Calcite, elongated rhombs                                           | Observational, ammonium carbonate method, modeling suggests tertiary structure clumps anionic sections of molecule                                                  | 21    |
| <b>Synthetic</b>                                                     |                                                                     |                                                                                                                                                                     |       |
| PSS <sup>4</sup>                                                     | Calcite, 3-7.5% of total crystallites oriented onto substrate       | Observational, ammonium carbonate method, % calcite formed correlates with degree of sulfation (to a plateau)                                                       | 16    |
| PSS <sup>4</sup> and poly(aspartic acid)                             | Calcite, 2-25% of total crystallites oriented onto substrate        | Observational, ammonium carbonate method, % affected by exposure time                                                                                               |       |
| PSS <sup>4</sup>                                                     | ACC <sup>1</sup>                                                    | Observational, gas diffusion method, nucleation inhibited in bulk solution                                                                                          | 22    |
| PSS <sup>4</sup>                                                     | ACC <sup>1</sup> that transforms to round, porous calcite           | Observational, gas diffusion method, crystals get rounder as PSS <sup>3</sup> or Ca <sup>2+</sup> inc.                                                              | 23,24 |
| PSS <sup>4</sup> (low conc.)                                         | Calcite, rhomb and vaterite, spherical, heterogeneous particle size | Observational, direct mixing, suggest sulfonate groups block CO <sub>3</sub> <sup>2-</sup> binding sites, induction time much higher than control at low saturation | 25,26 |
| PSS <sup>4</sup> (high conc.)                                        | Vaterite, spherical and homogenous particle size                    |                                                                                                                                                                     |       |
| PSS <sup>4</sup>                                                     | Vaterite, spheres                                                   | Observational, direct mixing, spheres composed of nanocrystals                                                                                                      | 27    |
| PSS <sup>4</sup>                                                     | Calcite, spherical                                                  | Observational, ammonium carbonate method, later added ethanol to mixture which inc. vaterite content                                                                | 28    |

|                                                               |                                                                        |                                                                                                                                                                                                                                   |    |
|---------------------------------------------------------------|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| PSS <sup>4</sup>                                              | Vaterite, smooth spheres                                               | Observational, direct mixing, suggest PSS <sup>4</sup> complexes Ca <sup>2+</sup> to block sites and repel CO <sub>3</sub> <sup>2-</sup>                                                                                          | 29 |
| PSS <sup>4</sup>                                              | Calcite/ vaterite mixture, spheres                                     | Observational, direct mixing, size of spheres increased with CaCO <sub>3</sub> conc. until threshold past which rhombs formed/ no PSS <sup>4</sup> effect                                                                         | 30 |
| PSS <sup>4</sup>                                              | Vaterite, uniform spheres                                              | Observational, direct mixing, suggest PSS <sup>4</sup> reduces driving force through interfacial energy and phase-stabilizes vaterite. High Ca <sup>2+</sup> :CO <sub>3</sub> <sup>2-</sup> ratio promoted vaterite more than low | 31 |
| PSS <sup>4</sup>                                              | Vaterite, 82%<br>Calcite, 18%                                          | Observational, rapid mixing, cauliflower shape until critical concentration, then spherical morphology                                                                                                                            | 32 |
| PVS <sup>5</sup>                                              | Vaterite, 86%<br>Calcite, 14%                                          |                                                                                                                                                                                                                                   |    |
| PSS <sup>4</sup> and polystyrene-polyethylene oxide copolymer | Vaterite, rough spheres                                                | Observational, direct mixing, copolymer forms micelles that nuclei form within leading to competition between complexation and adsorption                                                                                         | 29 |
| PSS <sup>4</sup> + CTAB <sup>6</sup> (low conc.)              | Calcite, rough spheres                                                 | Observational, direct mixing, hypothesize CTAB <sup>6</sup> changes PSS <sup>4</sup> morphology, reducing effect                                                                                                                  |    |
| PSS <sup>4</sup> + CTAB <sup>6</sup> (high conc.)             | Calcite, aggregated and faceted rhombs                                 | Observational, direct mixing, Size of spheres increased with CaCO <sub>3</sub> conc. until threshold past which rhombs formed/ no PSS <sup>4</sup> effect                                                                         |    |
| PSS <sup>4</sup> and PolyDADMAC <sup>7</sup>                  | Vaterite majority with some calcite and aragonite                      | Observational, direct mixing, cauliflower shape until critical concentration, then spherical morphology, % aragonite increased with concentration polyDADMAC <sup>7</sup>                                                         | 32 |
| PVS <sup>5</sup> and PolyDADMAC <sup>7</sup>                  | Vaterite majority with some calcite and aragonite                      |                                                                                                                                                                                                                                   |    |
| PVS <sup>5</sup>                                              | Vaterite, spheres                                                      | Observational, direct mixing, suggest PVS <sup>5</sup> slows rate by sequestering Ca <sup>2+</sup> and stabilizes vaterite, increased aggregation at high PVS <sup>5</sup> conc.                                                  | 33 |
| PSS <sup>4</sup>                                              | Vaterite, uniform spheres                                              | Observational, direct mixing, crystals became angular with time in the presence of PAMPS <sup>8</sup>                                                                                                                             | 34 |
| PAMPS <sup>8</sup>                                            | Vaterite, aggregated 'near' spheres                                    |                                                                                                                                                                                                                                   |    |
| PAMPS <sup>8</sup> -co-PAA <sup>2</sup>                       | Vaterite, 93%<br>Calcite, 7%<br>spheres                                | Observational, direct mixing                                                                                                                                                                                                      | 6  |
| PAMPS <sup>8</sup> -co-PAA <sup>2</sup>                       | Vaterite, 92.5%<br>Calcite, 7.5%                                       | Observational, direct mixing, smaller, more numerous crystals as concentration of PAMPS <sup>8</sup> -co-PAA <sup>2</sup> increases                                                                                               | 35 |
| Sulfated PMS <sup>9</sup> in solution                         | Calcite, rounded rhombs                                                | Observational, gas diffusion method, c-axis alignment and increased deformation as concentration S-PMS <sup>9</sup>                                                                                                               | 36 |
| Sulfated PMS <sup>9</sup> film                                | Calcite, rounded rhombs, selective nucleation based on chain positions | Observational, gas diffusion method, no c-axis alignment                                                                                                                                                                          |    |

|                                                                                                           |                                                                               |                                                                                                                                                                                       |    |
|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| PAMPS <sup>8</sup>                                                                                        | Calcite and aragonite, oriented triangular and tubular                        | Observational, gas diffusion method                                                                                                                                                   | 37 |
| PAMPS <sup>8</sup> -co-acrylamide hydrogel                                                                | Calcite, aggregates and cuboctahedral                                         | Observational, direct mixing, poly(acrylamide) alone gives pseudo-octahedral aggregates                                                                                               | 38 |
| Polymer with gradient amine content                                                                       | Calcite, rhombs                                                               | Observational, gas diffusion method, plasma polymerization used to make gradient surface, pH not specified                                                                            | 39 |
| C <sub>11</sub> -SH surface assembled monolayer                                                           | Calcite                                                                       | Quantitative, direct mixing, flow-through method<br>$\gamma = 91 \text{ mJ m}^{-2}$                                                                                                   | 40 |
| C <sub>16</sub> -SH surface assembled monolayer                                                           | Calcite                                                                       | $\gamma = 86 \text{ mJ m}^{-2}$                                                                                                                                                       |    |
| <i>n</i> -octadecanoic acid monolayer (CH <sub>3</sub> (CH <sub>2</sub> ) <sub>16</sub> COOH)             | Calcite, (11.0) face                                                          | Observational, direct mixing, monolayers at interface of air-supersaturated solution, experiments also performed with Mg <sup>2+</sup> present in which aragonite precipitated        | 41 |
| <i>n</i> -eicosyl sulfate monolayer (CH <sub>3</sub> (CH <sub>2</sub> ) <sub>19</sub> OSO <sub>3</sub> H) | Calcite, trigonal pyramidal (00.1) face                                       |                                                                                                                                                                                       |    |
| Natural/ synthetic mixtures or composites                                                                 |                                                                               |                                                                                                                                                                                       |    |
| PSS <sup>4</sup> and proteins from <i>M. californianus</i> (bivalve)                                      | Calcite, 17.5% total crystallites oriented onto substrate                     | Observational, ammonium carbonate method, protein presence increased % calcite 3 to 17.5%, Suggest sulfonate and carboxylate cooperation                                              | 16 |
| PSS <sup>4</sup> and proteins from <i>M. californianus</i> (bivalve) without carboxylate                  | Calcite, 3-5.7% of total crystallites oriented onto substrate                 | Observational, ammonium carbonate method, carboxylate blocked by crosslinking                                                                                                         |    |
| Polymer with gradient amine content and Chondroitin sulfate                                               | Calcite, sphere or ellipsoid, much higher density where amine content is high | Observational, gas diffusion method, plasma polymerization to make gradient surface, pH not specified                                                                                 | 39 |
| Chitosan and PAMPS <sup>8</sup> -co-PAA <sup>2</sup>                                                      | Vaterite, 88%<br>Calcite, 12%<br>Rough spheres, aggregates                    | Observational, direct mixing, suggest chain rigidity influences morphology                                                                                                            | 6  |
| Chitosan and PAMPS <sup>8</sup> -co-PAA <sup>2</sup>                                                      | Vaterite, 88%<br>Calcite, 12%<br>Rough spheres                                | Observational, direct mixing, smaller, more numerous crystals as concentration of PAMPS <sup>8</sup> -co-PAA <sup>2</sup> increases, larger fraction of calcite when chitosan present | 35 |
| PVS <sup>5</sup> grafted chitosan                                                                         | All polymorphs, various morphologies (spherical, disk, prismatic, needle)     | Observational, gas diffusion method, CaSO <sub>4</sub> also present due to crosslinker releasing SO <sub>4</sub> <sup>2-</sup>                                                        | 42 |
| $\beta$ -cyclodextrin-maleic anhydride, PSS <sup>4</sup> terpolymer and chitosan                          | Aggregates and spherical granules, up to 96% inhibition                       | Observational, scale inhibition test, % inhibition increased as % monomer conversion to PSS <sup>4</sup> increased                                                                    | 43 |
| Cellulose-based woven material and PAMPS <sup>8</sup>                                                     | Calcite composite layer, crystallites 576 Å                                   | Observational, ammonium carbonate method, control produced thin calcite layer,                                                                                                        | 44 |

|                                                                                                                |                                             |                                                               |  |
|----------------------------------------------------------------------------------------------------------------|---------------------------------------------|---------------------------------------------------------------|--|
|                                                                                                                |                                             | while some larger crystals were present in composite layer(s) |  |
| Cellulose-based woven material and PAMPS <sup>8</sup> -co-PAA <sup>2</sup>                                     | Calcite composite layer, crystallites 163 Å |                                                               |  |
| Cellulose-based woven material with chitosan and PAMPS <sup>8</sup>                                            | Calcite composite layer, crystallites 979 Å |                                                               |  |
| Cellulose-based woven material with chitosan and PAMPS <sup>8</sup> -co-PAA <sup>2</sup>                       | Calcite composite layer, crystallites 984 Å |                                                               |  |
| Cellulose-based woven material with poly(allylamine hydrochloride) and PAMPS <sup>8</sup>                      | Calcite composite layer, crystallites 401 Å |                                                               |  |
| Cellulose-based woven material with poly(allylamine hydrochloride) and PAMPS <sup>8</sup> -co-PAA <sup>2</sup> | Calcite composite layer, crystallites 939 Å |                                                               |  |

<sup>1</sup>Amorphous calcium carbonate; <sup>2</sup>Polyacrylic acid; <sup>3</sup>Octocania proteoglycan from rat with chondroitin sulfate linkages; <sup>4</sup>Polystyrene sulfonate; <sup>5</sup>polyvinyl sulfate; <sup>6</sup>Cetyltrimethyl ammonium;

<sup>7</sup>Poly(diallyldimethylammonium chloride); <sup>8</sup>Poly(2-acrylamido-2-methyl-1-propanesulfonic acid);

<sup>9</sup>Polymethyl siloxane

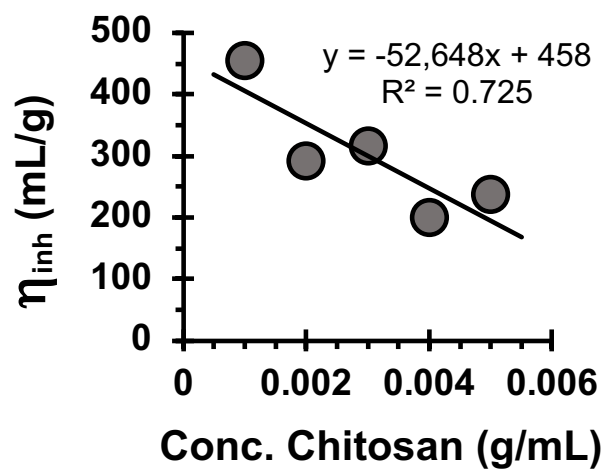
**Table S2:** Synthesis conditions for each sulfated chitosan material used in this study.

| <b>O-sulfation</b> |                     |                                           |                                         |                            |                   |                          |
|--------------------|---------------------|-------------------------------------------|-----------------------------------------|----------------------------|-------------------|--------------------------|
| <b>Sample</b>      | <b>Chitosan (g)</b> | <b>HSO<sub>3</sub>Cl (mL)</b>             | <b>HCOOH (mL)</b>                       | <b>DMF (mL)</b>            | <b>Temp. (°C)</b> | <b>Reaction time (h)</b> |
| OSC 0.23           | 0.501               | 2                                         | 15                                      | 95                         | 50                | 3                        |
| OSC 0.42           | 1.020               | 7                                         | 20                                      | 200                        | 50                | 3                        |
| OSC 0.77           | 0.504               | 5                                         | 20                                      | 125                        | 50                | 3                        |
| <b>N-sulfation</b> |                     |                                           |                                         |                            |                   |                          |
| <b>Sample</b>      | <b>Chitosan (g)</b> | <b>Me<sub>3</sub>N-SO<sub>3</sub> (g)</b> | <b>Na<sub>2</sub>CO<sub>3</sub> (g)</b> | <b>H<sub>2</sub>O (mL)</b> | <b>Temp. (°C)</b> | <b>Reaction time (h)</b> |
| NSC 0.13 a         | 0.3506              | 0.8753                                    | 0.4547                                  | 60                         | 50                | 4                        |
| NSC 0.13 b         | 0.815               | 2.0495                                    | 1.2015                                  | 150                        | 50                | 3                        |
| NSC 0.28           | 0.5099              | 1.3597                                    | 0.8683                                  | 50                         | 60                | 12                       |
| NSC 0.47           | 0.5080              | 1.3690                                    | 0.8680                                  | 50                         | 60                | 24                       |

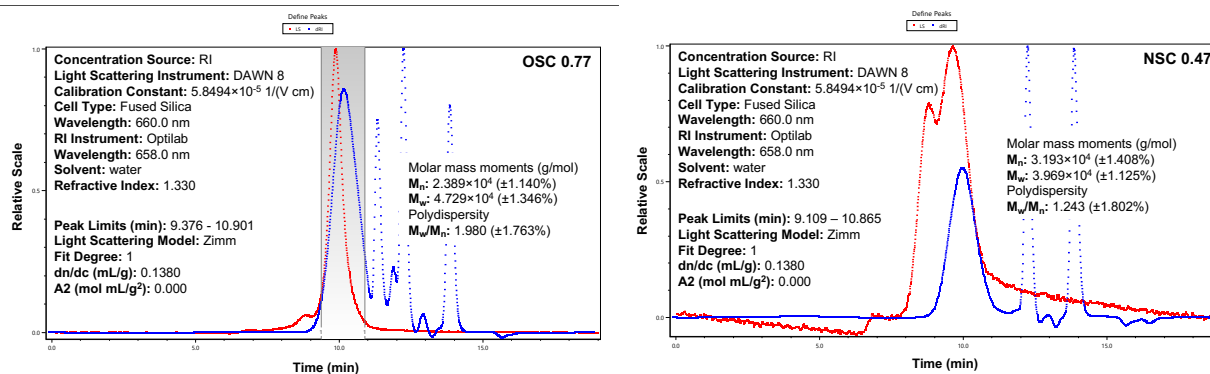
**Table S3:** Solutions for nucleation experiments were prepared based upon the concentrations required to achieve a desired supersaturation and a near-one activity ratio upon mixing. The compositions targeted a reaction pH of 9.7 but actual experimental pH values ranged from 9.7 – 10.0 upon mixing. Thus,  $\sigma$  varied slightly based on the pH of each mixed solution.

| Desired $\sigma$<br>upon mixing | mM $\text{CaCl}_2$<br>(pre-mixing) | mM $\text{NaHCO}_3$<br>(pre-mixing) | $a_{\text{Ca}^{2+}} : a_{\text{CO}_3^{2-}}$<br>upon mixing |
|---------------------------------|------------------------------------|-------------------------------------|------------------------------------------------------------|
| 4.67                            | 3.00                               | 7.50                                | 1.00                                                       |
| 4.87                            | 3.50                               | 10.00                               | 1.01                                                       |
| 5.04                            | 4.00                               | 11.00                               | 1.03                                                       |
| 5.26                            | 4.80                               | 12.50                               | 1.05                                                       |
| 5.43                            | 5.40                               | 14.00                               | 1.03                                                       |
| 5.66                            | 6.50                               | 14.25                               | 1.02                                                       |

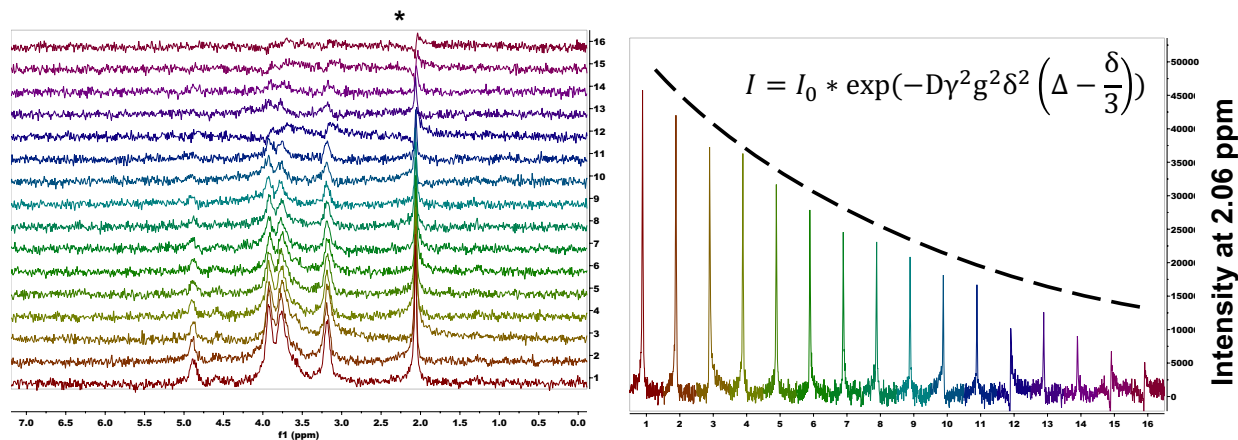
**Figure S1:** The intrinsic viscosity and molecular weight of the starting chitosan material (Chitosan A) were estimated by determining the inherent viscosity ( $\eta_{inh}$ ) for a series of concentrations at ambient temperature in pH 4.5 DI H<sub>2</sub>O/ acetic acid.



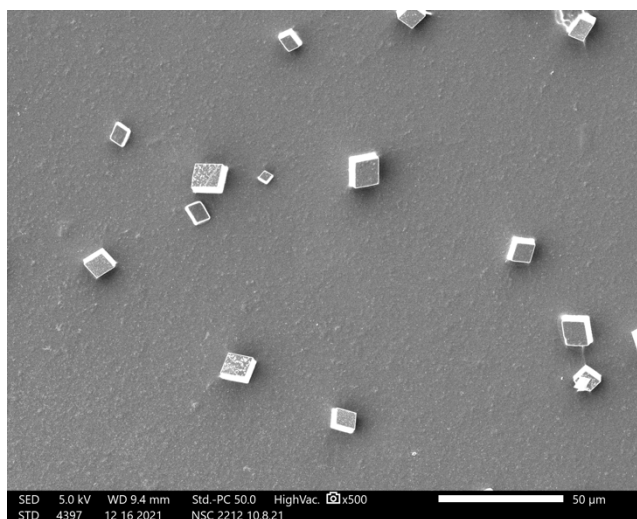
**Figure S2:** Aqueous SEC chromatograms for OSC 0.77 (**left**) and NSC 0.47 (**right**) collected at 50 °C in pH 3.0 DI H<sub>2</sub>O/ acetic acid.



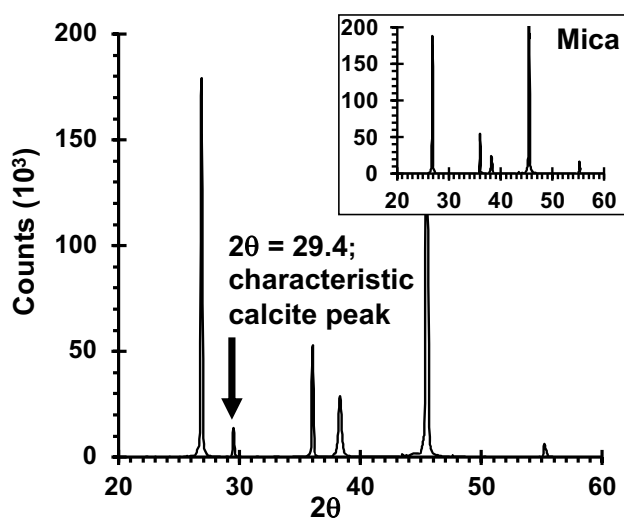
**Figure S3:** Example of data from NMR diffusometry measurements of the diffusion coefficient for chitosan materials. **(Left)**  $^1\text{H}$  NMR spectra of chitosan over various gradient strengths collected at 25 °C on a 9.4 T Bruker Avance III. Water signal is removed. **(Right)** The intensity of the peak corresponding to the methyl protons of the acetyl group (\*2.06 ppm for chitosan) over experiment number to display an exponential dependence (Mnova). The intensity ( $I$ ) vs gradient strength ( $g$ ) data is fit with the Stejskal-Tanner equation (using Topspin) to extract the diffusion coefficient ( $D$ ) using the equation (inset) where  $I_0$  is the signal intensity in the absence of a gradient,  $\gamma$  is the gyromagnetic ratio ( $\gamma_{1\text{H}} = 26.75 \times 10^7 \text{ rad T}^{-1} \text{ s}^{-1}$ ),  $\delta$  is the gradient pulse length (2 ms), and  $\Delta$  is the diffusion time (20 – 30 ms). The average error for diffusion measurements is < 10%.



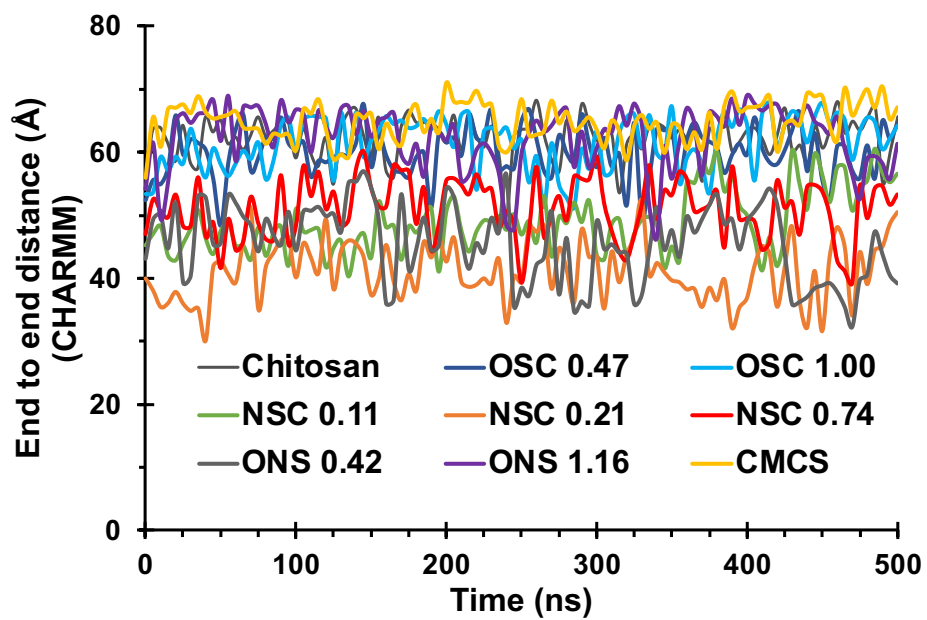
**Figure S4:** SEM image of a typical surface at the end of a nucleation experiment with rhombohedral calcite crystals on a relatively uniform polysaccharide substrate. The sample shown is from an experiment with *N*-sulfated chitosan surface with  $DS(SO_3^-) = 0.28$ . Magnification = 500x



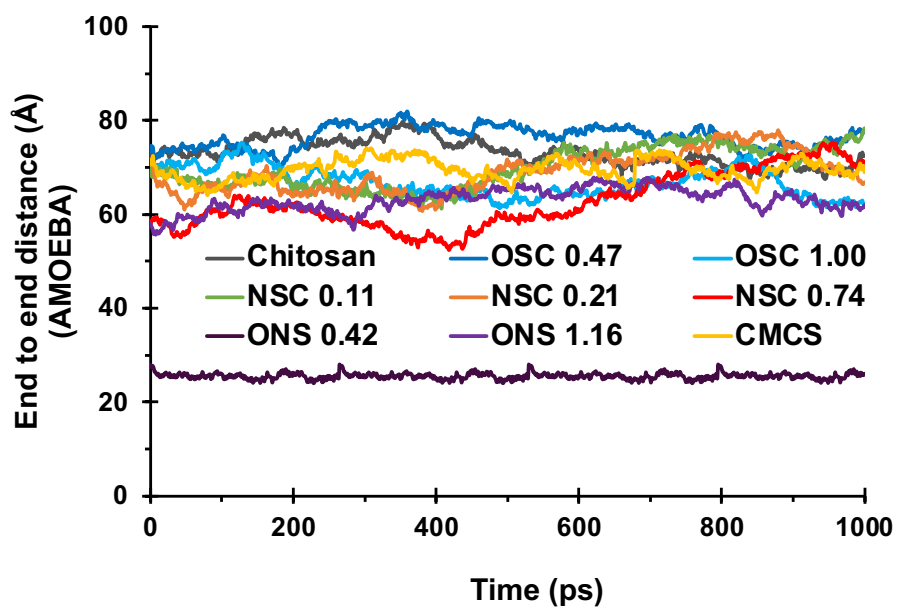
**Figure S5:** A representative X-ray diffractogram shows the calcite polymorph formed during nucleation onto an *N*-sulfated chitosan surface,  $DS(SO_3^-) = 0.28$ .



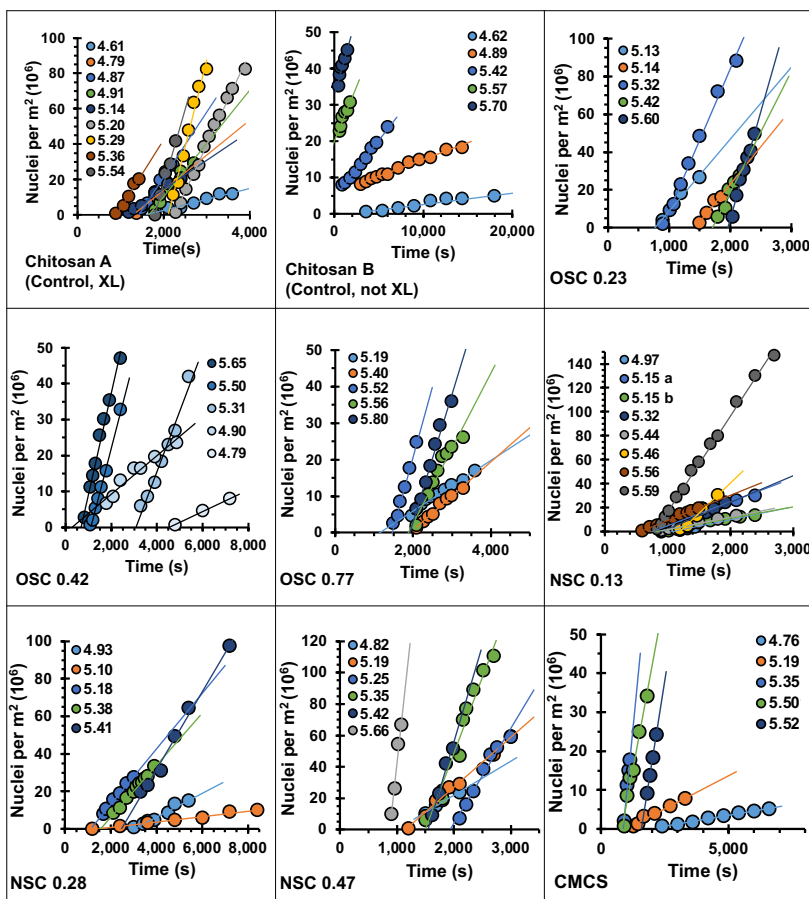
**Figure S6:** End-to-end distances, calculated using the CHARMM MD trajectories, show no major changes over the course of the simulation, suggesting an equilibrium conformation was adopted.



**Figure S7:** End-to-end distances, calculated from the AMOEBA MD trajectories, show all materials maintain an approximately constant conformation throughout the simulation. The composition ONS 0.42 exhibits a significantly lower end-to-end distance than other polymers, indicating a more intertwined/tangled orientation.



**Figure S8:** Nuclei vs time data for investigated materials. To obtain flux,  $J$ , the crystallite count is converted from number per viewing window to SI units (see statement of assumptions in manuscript).

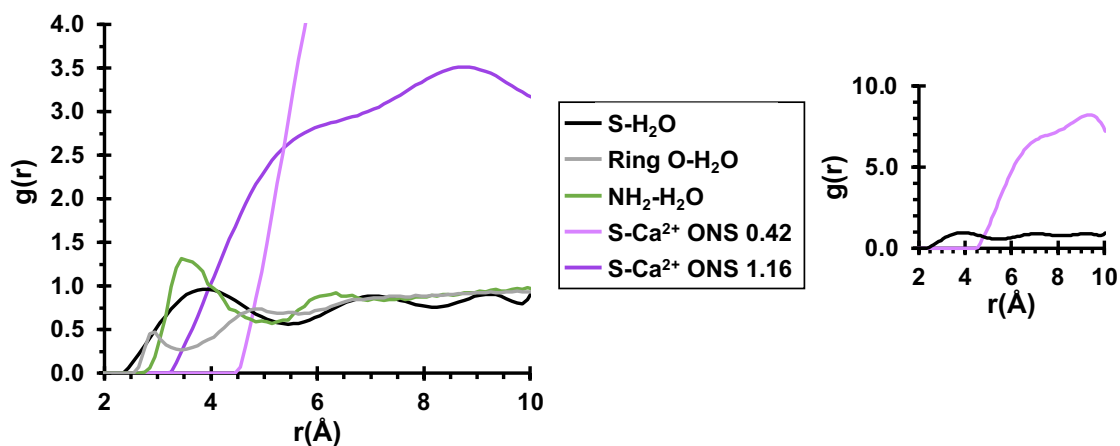


**Table S4:** Values of B and ln(A) for each material and the corresponding estimates of  $\gamma_{\text{net}}$  values that are represented in **Figures 3 - 7**.

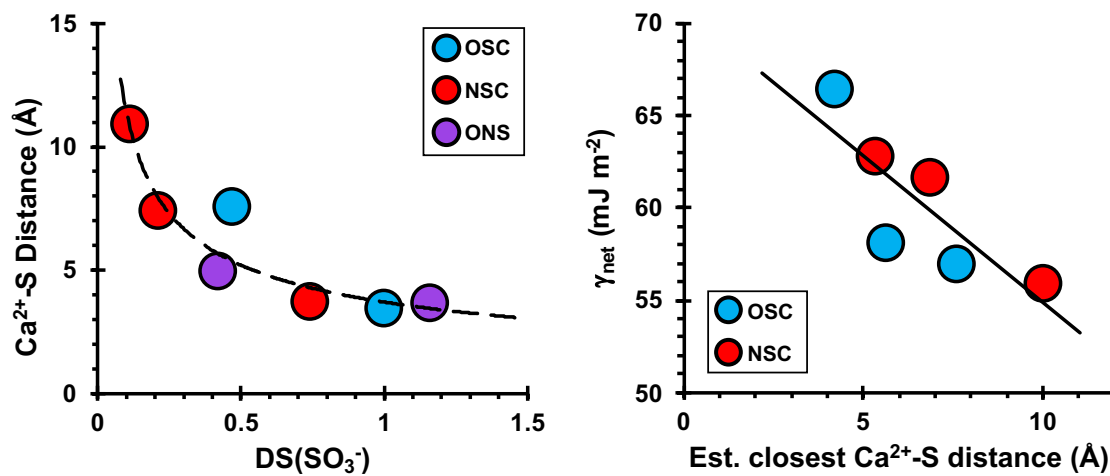
| Material                           | DS(Anionic group) | B $\pm$ SE <sup>1</sup> | ln(A) $\pm$ SE <sup>1</sup> | $\gamma_{\text{net}}$ (mJ m <sup>-2</sup> ) |
|------------------------------------|-------------------|-------------------------|-----------------------------|---------------------------------------------|
| <b>This study</b>                  |                   |                         |                             |                                             |
| Chitosan A                         | 0                 | 148 $\pm$ 74            | 16.2 $\pm$ 2.6              | 57                                          |
| Chitosan B                         | 0                 | 186 $\pm$ 11            | 14.6 $\pm$ 0.4              | 59                                          |
| OSC 0.23                           | 0.23              | 171 $\pm$ 30            | 17.1 $\pm$ 1.1              | 57                                          |
| OSC 0.42                           | 0.42              | 182 $\pm$ 10            | 16.0 $\pm$ 0.4              | 58                                          |
| OSC 0.77                           | 0.77              | 273 $\pm$ 86            | 17.8 $\pm$ 2.9              | 66                                          |
| NSC 0.13                           | 0.13              | 162 $\pm$ 81            | 15.7 $\pm$ 2.9              | 56                                          |
| NSC 0.28                           | 0.28              | 217 $\pm$ 172           | 17.1 $\pm$ 6.4              | 62                                          |
| NSC 0.47                           | 0.47              | 230 $\pm$ 47            | 19.5 $\pm$ 1.7              | 63                                          |
| CMCS                               | 1.6               | 346 $\pm$ 99            | 22.1 $\pm$ 3.6              | 72                                          |
| <b>Giuffre et al., PNAS (2014)</b> |                   |                         |                             |                                             |
| Chitosan (not XL)                  | 0                 | 121 $\pm$ 22            | not reported                | 51                                          |
| Hyaluronic acid                    | 0.5               | 253 $\pm$ 58            | not reported                | 65                                          |
| Low G-block alginate               | 1                 | 294 $\pm$ 11            | not reported                | 72                                          |
| High G-block alginate              | 1                 | 390 $\pm$ 45            | not reported                | 76                                          |
| DeN- sulfated heparin              | 1.3               | 376 $\pm$ 60            | not reported                | 75                                          |

<sup>1</sup>Standard error

**Figure S9:** (left) Simulations of the two O- and N-sulfated chitosan (ONS) materials also predict  $\text{Ca}^{2+}$  is solvent-separated from the sulfate groups.  $\text{Ca}^{2+}$  is concentrated in the low water density region at  $\approx 5 \text{ \AA}$  with a secondary population of  $\text{Ca}^{2+}$  at  $\approx 8 \text{ \AA}$  (purple). (right) Full S- $\text{Ca}^{2+}$  RDF for ONS 0.42. Atoms in this system are more concentrated near the polymer due to its tangled conformation (significantly lower end-to-end distance, **Figure S7**).



**Figure S10: (left)** Smallest distance between a sulfated group and  $\text{Ca}^{2+}$  is inversely correlated with  $\text{DS}(\text{SO}_3^-)$  and declines to a near-constant value of  $\approx 3.5 \text{ \AA}$ , independent of sulfate position. **(right)** The closest  $\text{Ca}^{2+}$ -S distances are inversely correlated with  $\gamma_{\text{net}}$  values determined from experiment.



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